

2024 Paper 2 Questions by Topic

- 1 (a) Table 1.1 lists some SI quantities. Complete the table by indicating with a tick (✓) which rows are SI base quantities.

current and mass only ticked	A1	Allow other methods to indicate current and mass e.g. 'x' or underlining, provided no other methods are seen. Ignore units in all boxes
	quantity	base quantity
	current	
	energy	
	force	
	mass	

[1]

- (b) Use the definition of power to determine its SI base units.

(power =) work (done) / time Units of power = J s^{-1} = $\text{kg m}^2 \text{s}^{-2} / \text{s}$ = $\text{kg m}^2 \text{s}^{-3}$	C1 A1	Allow equivalent expressions e.g. Force $\times$ velocity, current ² $\times$ resistance, intensity $\times$ area Allow in symbols e.g. W/t, (Δ)E/t, Fd/t, IA or e.g. units A $\times$ V J s ⁻¹ or J / s implies first mark Allow 'WD'/'Wd' to mean 'work done' BOD
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SI base units [2]

- (c) A light meter is used to measure the intensity of light in a classroom. Daylight is incident normally on the sensor of the meter. The sensor has an area of 2.2cm^2 . The reading on the meter is 950W m^{-2} .

Calculate the power of the daylight incident on the sensor.

power = intensity $\times$ area = $950 \times 2.2 \times 10^{-4}$ = 0.21W	C1 C1 A1	Allow $P = I \times A$. Allow any subject Treat incorrect or no conversion of cm^2 to m^2 as POT and Allow ECF for A1 mark
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power = W [3]

[Total: 6]

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1 The drag force F_D acting on an object falling through air is given by

$$F_D = \frac{1}{2} C \rho A v^2$$

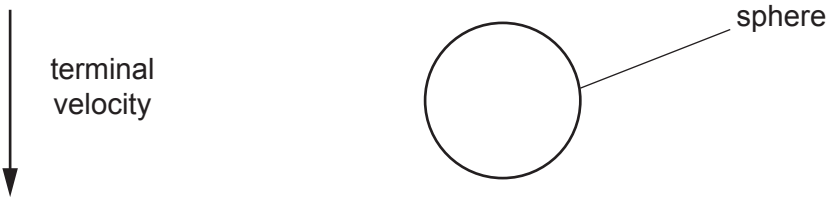
where A is the cross-sectional area of the object,
 v is the velocity of the object in the air,
 ρ is the density of the air and
 C is a constant called the drag coefficient.

(a) Use SI base units to show that the drag coefficient has no units.

1(a)	units of F_D : kg m s^{-2}	M1
	units of ρ : kg m^{-3} and units of A : m^2 and units of v : m s^{-1} or units of v^2 : $\text{m}^2 \text{s}^{-2}$	M1
	$\text{kg m s}^{-2} = C \text{ kg m s}^{-2}$ and comment '(so) C has no units' / unit terms cancelled or $C = \text{kg m s}^{-2} / (\text{kg m}^{-3} \text{ m}^2 \text{ m}^2 \text{s}^{-2})$ and comment '(so) C has no units' / unit terms cancelled	A1

[3]

(b) Fig. 1.1 shows a sphere falling at terminal velocity in air.



1(b)	one arrow vertically downward labelled weight to within 10° of the vertical	B1
	one arrow vertically upwards labelled drag / drag force / F_D / air resistance / viscous force to within 10° of the vertical	B1

Assume that the upthrust on the sphere is negligible.

On Fig. 1.1, draw and label arrows to show the directions of the **two** forces acting on the sphere. [2]

(c) The mass of the sphere is 49 g.

Calculate the drag force F_D acting on the sphere.

1(c)	(at terminal velocity) $F_D = mg$	C1
	$F_D = 0.049 \times 9.81$	A1
	$= 0.48 \text{ N}$	

$F_D = \dots\dots\dots \text{ N}$ [2]

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- (d) The sphere is falling in air at a terminal velocity of 25 in SI base units.
The density of the air is 1.2 in SI base units.
The diameter of the sphere is 0.060 in SI base units.

Use your answer in (c) to calculate the drag coefficient C for the sphere.

1(d)	$area = \pi \times (0.060 / 2)^2$	C1
	$0.48 = \frac{1}{2} \times C \times 1.2 \times \pi \times (0.060 / 2)^2 \times 25^2$	C1
	$C = 0.45$	A1

$C = \dots\dots\dots$ [3]

[Total: 10]

1 (a) The list below shows some SI quantities.

Underline the quantity that is **not** an SI base quantity.

1(a)	charge underlined (and no others)	B1
	charge current length time	[1]

(b) A square solar panel with sides of length 1300 mm is shown in Fig. 1.1.

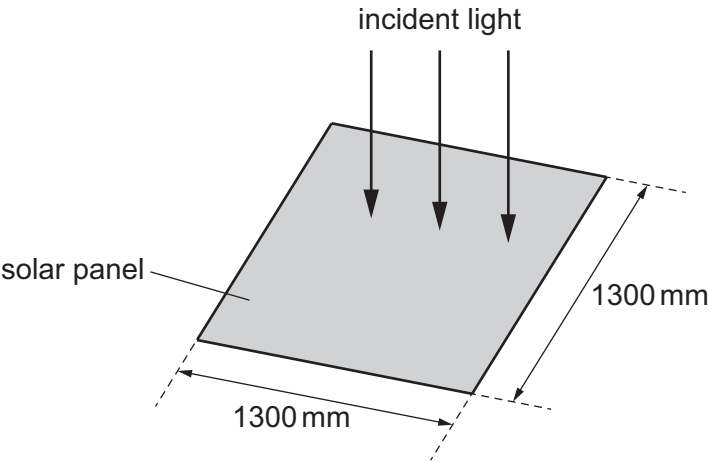


Fig. 1.1 (not to scale)

Light is incident normally on the solar panel.

(i) The power of the light incident on the solar panel is 750 W.

Calculate the intensity of the light.

1(b)(i)	$I = P / A$	C1
	$= 750 / (1300 \times 10^{-3})^2$	C1
	$= 440 \text{ W m}^{-2}$	A1

intensity = W m^{-2} [3]

(ii) The percentage uncertainty in the incident power is $\pm 3\%$.
The uncertainty in the length of each side is $\pm 5 \text{ mm}$.

Calculate the percentage uncertainty in the intensity of the light.

1(b)(ii)	percentage uncertainty $= 3 + 2 \times (5 / 1300) \times 100$	C1
	$= 3 + 2 \times 0.38$	A1
	$= (\pm) 4\%$	

percentage uncertainty = % [2]

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(iii) The useful power output of the solar panel is 160 W.

Calculate the percentage efficiency of the solar panel.

1(b)(iii)	efficiency = useful output power / total input power = (160 / 750) × 100 = 21%	A1
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efficiency = % [1]

(iv) Another square solar panel is placed so that light of the same intensity is incident normally on it. The new panel has shorter sides than the original panel. The new panel has the same power output as the original panel.

State and explain whether the efficiency of the new panel is greater than, less than or the same as the efficiency of the original panel.

.....

1(b)(iv)	area (of the new panel) is less	B1
	input power (of the new panel) is less (than the input power of the original panel) (as intensity is constant)	B1
	(useful power output is unchanged so) efficiency is greater (than the original panel)	B1

.....

..... [3]

[Total: 10]

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1 The drag force F_D acting on a sphere falling through a liquid is given by

$$F_D = 6\pi\eta r v$$

where r is the radius of the sphere,
 v is the speed of the sphere in the liquid and
 η is a property of the liquid called the viscosity.

(a) Show that the SI base units of viscosity are $\text{kg m}^{-1} \text{s}^{-1}$.

1(a)	units of F : kg m s^{-2}	C1
	units of r : m and units of v : m s^{-1}	A1
	units of η : $\text{kg m s}^{-2} / (\text{m} \times \text{m s}^{-1}) = \text{kg m}^{-1} \text{s}^{-1}$	

[2]

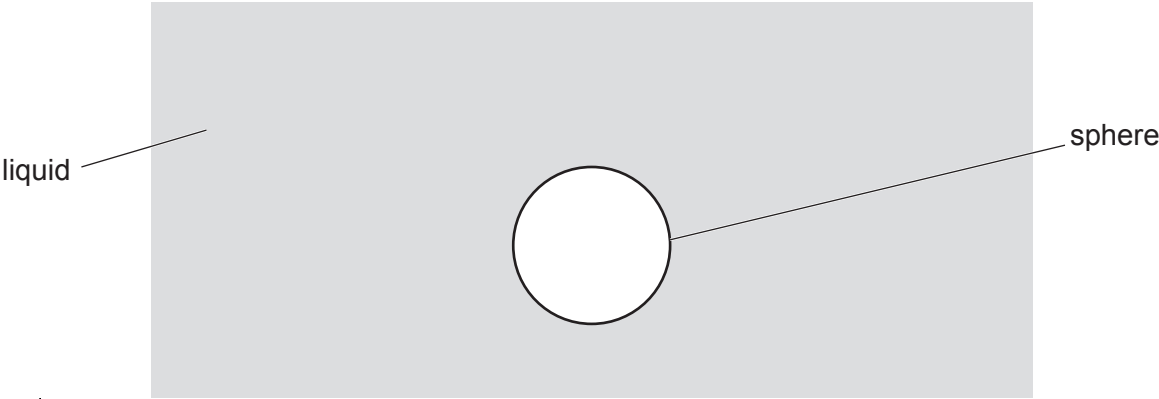
(b) The sphere has a radius of 3.0 cm and is falling vertically downwards at a terminal velocity of 2.0 m s^{-1} through the liquid. The drag force acting on the sphere is 0.096 N.

Calculate the viscosity of the liquid.

1(b)	viscosity = $0.096 / (6 \times \pi \times 0.03 \times 2.0)$	C1
	= $0.085 \text{ kg m}^{-1} \text{s}^{-1}$	A1

viscosity = $\text{kg m}^{-1} \text{s}^{-1}$ [2]

(c) The sphere is shown in Fig. 1.1.



1(c)	one arrow vertically downwards labelled weight / W	B1
	arrow(s) vertically upwards labelled U / upthrust and drag / F_D / viscous force	B1

Fig. 1.1

On Fig. 1.1, draw and label arrows to represent the directions of the **three** forces acting on the sphere as it falls at terminal velocity. [2]

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(d) (i) The density of the liquid is 920 kg m^{-3} .

Show that the upthrust acting on the sphere is 1.0 N .

1(d)(i)	$V = (4 / 3) \pi r^3$	C1
	$\text{upthrust} = (4 / 3) \times \pi \times 0.03^3 \times 920 \times 9.81 = 1.0\text{ N}$	A1

[2]

(ii) Calculate the mass of the sphere.

1(d)(ii)	$\text{weight} = 1.0 + 0.096 (= 1.096\text{ N})$	C1
	$m = 1.096 / 9.81$	A1
	$= 0.11\text{ kg}$	

mass = kg [2]

[Total: 10]

1 (a) Define density.

mass per unit volume	B1
.....	[1]

(b) Fig. 1.1 shows a cuboidal glass block.

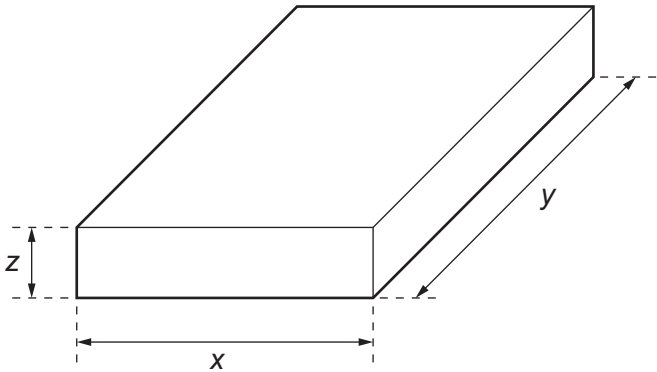


Fig. 1.1 (not to scale)

A student measures the mass m of the block and the side lengths x , y and z . The measurements are shown in Table 1.1.

Table 1.1

quantity	measurement
m	$(0.243 \pm 0.001)\text{ kg}$
x	$(5.41 \pm 0.01)\text{ cm}$
y	$(11.09 \pm 0.01)\text{ cm}$
z	$(1.62 \pm 0.01)\text{ cm}$

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(i) Determine the density of the glass.

density = $0.243 / (0.0541 \times 0.1109 \times 0.0162)$	C1
= 2500 kg m^{-3}	A1

density = kg m^{-3} [2]

(ii) Calculate the percentage uncertainty in the density.

$(0.001 / 0.243) \text{ or } (0.01 / 5.41) \text{ or } (0.01 / 11.09) \text{ or } (0.01 / 1.62)$	C1
$(\Delta\rho / \rho) = (\Delta m / m) + (\Delta x / x) + (\Delta y / y) + (\Delta z / z)$ $= (0.001 / 0.243) + (0.01 / 5.41) + (0.01 / 11.09) + (0.01 / 1.62)$ $(= 0.013)$	C1
percentage uncertainty in $\rho = 0.013 \times 100$ $= 1.3\%$ (allow 1 s.f. answer of 1%)	A1

percentage uncertainty =% [3]

(c) The true value of the density of the glass is different from the answer in (b)(i) because of a systematic error in the measurements.

Suggest **one** possible cause of this systematic error.

zero error (on calipers / balance) or incorrect calibration (of calipers / balance)	B1
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..... [1]

[Total: 7]

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1 (a) State what is meant by a vector quantity.

a quantity with magnitude and direction	B1
..... [1]	

(b) A sphere falls vertically through a liquid that has density 830 kg m^{-3} . The sphere has radius r and constant velocity v , as shown in Fig. 1.1.

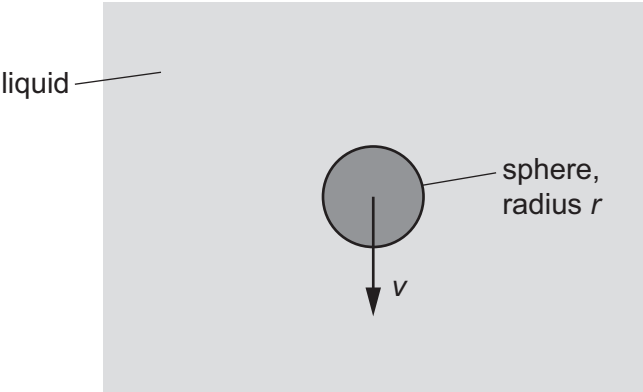


Fig. 1.1

(i) The drag force D acting on the sphere is given by

$$D = 6\pi r\eta v$$

where η is a property of the liquid.

Determine the SI base units of η .

SI base units of D : kg m s^{-2}	C1
SI base units of r : m and v : m s^{-1}	C1
base units of η : $\text{kg m s}^{-2} / (\text{m} \times \text{m s}^{-1})$ $= \text{kg m}^{-1} \text{s}^{-1}$	A1

SI base units [3]

(ii) State an equation showing the relationship between the magnitudes of the weight W , drag force D and upthrust U acting on the sphere.

..... [1]

$W = U + D$	A1
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(iii) The volume of the sphere is 4.6 cm^3 . The drag force D is 0.32 N .

Calculate the weight of the sphere.

$U = 830 \times 9.81 \times 4.6 \times (10^{-2})^3$ (= 0.037 N)	C1
$W = 0.037 + 0.32$ = 0.36 N	A1

weight = N [2]

[Total: 7]